

Leon O. Chua

其他

基本信息

姓名: Leon O. Chua

所属院校: 加州大学-伯克利

地区: United States

年龄范围: 50岁以上

海外经历: 有

荣誉称号

★ Member of Academia Europaea

★ IEEE Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

电子与计算机工程

系统科学与工程

电子科学与技术

人工智能

微电子科学与工程

应用物理学

物理学

集成电路设计与集成系统

研究领域

系统动力学与仿真

深度学习与神经网络

混沌与分形系统

纳米电子与先进工艺

电路与系统设计

微电子器件与制造

半导体器件物理与工艺

微纳电子与纳米技术

统计物理与复杂系统

存储器设计与架构

+2

个人简介

Leon O. Chua, a distinguished professor of Electrical Engineering at UC Berkeley, holds a Ph.D. from the University of Illinois, Urbana-Champaign. His

research focuses on biosystems, computational biology, control systems, nanoelectronics, and nonlinear dynamics. Chua is the recipient of the 2005 Gustav Kirchhoff Award and the IEEE Neural Networks Pioneer Award, among numerous other accolades. He has been elected a Fellow of the IEEE and a foreign member of the European Academy of Sciences. His contributions to the field of electronic circuits and systems have been widely recognized with multiple patents and honorary doctorates.

工作经历

Purdue University - Assistant Professor	1964 - 1967
Purdue University - Associate Professor	1967 - 1970
UC Berkeley	1970 - 至今

教育经历

University of Illinois, Urbana-Champaign - Ph.D.	- 1964
Massachusetts Institute of Technology - M.S.	- 1961
Mapúa Institute of Technology - B.S.	- 1959

近年发表的论文

● 电子与计算机工程 ● 系统科学与工程 ● 电子科学与技术 ● 人工智能 ● 微电子科学与工程

1999

20020052008201020122015201820202220232024

19992024

Modelling Thermal Crosstalk on Temperature Driven Memristors

2024 13TH INTERNATIONAL CONFERENCE ON MODERN CIRCUITS AND SYSTEMS TECHNOLOGIES, MOCAST 2024

Bush-Espinosa, Lydia; de Benito, Carol; Stavrinides, Stavros G.; Picos, Rodrigo; Chua, Leon O.

被引用次数: 02024

Slow Invariant Manifolds of Memristor-Based Chaotic Circuits

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Ginoux, Jean-Marc; Meucci, Riccardo; Chen, Guanrong; Chua, Leon O.

被引用次数: 2

2024

Edge of Chaos Theory Sheds Light Into the All-to-None Phenomenon in Neurons-Part I: On the Fundamental Role of the Sodium Ion Channel

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ascoli, Alon; Demirkol, Ahmet Samil; Tetzlaff, Ronald; Chua, Leon O.

被引用次数: 6

2024

Poor Man's Memristor: Chua Corsage Memristor

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jin, Peipei; Wang, Guangyi; Liang, Yan; Chen, Long; lu, Herbert Ho-Ching; Chua, Leon O.

被引用次数: 4

2023

Dynamical structural phase transition model of the electrical behavior of NiTi

PHYSICA B-CONDENSED MATTER

Antoniades, Ioannis P.; Stavriniades, Stavros G.; Picos, Rodrigo; Haniias, Michael P.; Hatzikraniotis, Euripides; Georgiou, Julius; Chua, Leon O.

被引用次数: 2

2023

Experimental observation of chaotic hysteresis in Chua's circuit driven by slow voltage forcing

CHAOS SOLITONS & FRACTALS

Gomes, lacyel; Korneta, Wojciech; Stavriniades, Stavros G.; Picos, Rodrigo; Chua, Leon O.

被引用次数: 18

2023

Research Progress on Memristor: From Synapses to Computing Systems

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Yang, Xiaoxuan; Taylor, Brady; Wu, Ailong; Chen, Yiran; Chua, Leon O.

被引用次数: 58

2022

Experimental validation of state equations and dynamic route maps for phase change memristive devices

SCIENTIFIC REPORTS

Marrone, Francesco; Secco, Jacopo; Kersting, Benedikt; Le Gallo, Manuel; Corinto, Fernando; Sebastian, Abu; Chua, Leon O.

被引用次数: 5

2022

Empirical Characterization of ReRAM Devices Using Memory Maps and a Dynamic Route Map

ELECTRONICS

Picos, Rodrigo; Stavriniades, Stavros G.; Al Chawa, Mohamad Moner; de Benito, Carola; Duenas, Salvador; Castan, Helena; Hatzikraniotis, Euripides; Chua, Leon O.

被引用次数: 1

2022

Using Self-Heating Resistors as a Case Study for Memristor Compact Modeling

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Picos, Rodrigo; Al Chawa, Mohamad Moner; De Benito, Carola; Stavrinides, Stavros G.; Chua, Leon O.

被引用次数: 1

2022

Edge of Chaos Explains Prigogines Instability of the Homogeneous

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Ascoli, Alon; Demirkol, Ahmet S.; Tetzlaff, Ronald; Chua, Leon O.

被引用次数: 9

2022

On the chaotic nature of random telegraph noise in unipolar RRAM memristor devices

CHAOS SOLITONS & FRACTALS

Stavrinides, Stavros G.; Hantias, Michael P.; Gonzalez, Mireia B.; Campabadal, Francesca; Contoyiannis, Yiannis; Potirakis, Stelios M.; Chawa, Mohamad Moner Al; de Benito, Carol; Tetzlaff, Ronald; Picos, Rodrigo; Chua, Leon O.

被引用次数: 5

2022

A Compact and Continuous Reformulation of the Strachan TaOx Memristor Model With Improved Numerical Stability

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Demirkol, Ahmet S.; Ascoli, Alon; Messaris, Ioannis; Al Chawa, Mohamad Moner; Tetzlaff, Ronald; Chua, Leon O.

被引用次数: 6

2022

On the Thermal Models for Resistive Random Access Memory Circuit Simulation

NANOMATERIALS

Roldan, Juan B.; Gonzalez-Cordero, Gerardo; Picos, Rodrigo; Miranda, Enrique; Palumbo, Felix; Jimenez-Molinos, Francisco; Moreno, Enrique; Maldonado, David; Baldoma, Santiago B.; Moner Al Chawa, Mohamad; de Benito, Carol; Stavrinides, Stavros G.; Sune, Jordi; Chua, Leon O.

被引用次数: 43

2021

All-Optically Controlled Memristor for Optoelectronic Neuromorphic Computing

ADVANCED FUNCTIONAL MATERIALS

Hu, Lingxiang; Yang, Jing; Wang, Jingrui; Cheng, Peihong; Chua, Leon O.; Zhuge, Fei

被引用次数: 161

2021

NbO₂-Mott Memristor: A Circuit-Theoretic Investigation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Messaris, Ioannis; Brown, Timothy D.; Demirkol, Ahmet S.; Ascoli, Alon; Al Chawa, M. Moner; Williams, R. Stanley; Tetzlaff, Ronald; Chua, Leon O.

被引用次数: 30

2021

Analog Neural Computing With Super-Resolution Memristor Crossbars

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

James, Alex P.; Chua, Leon O.

被引用次数: 25

2021

On Local Activity and Edge of Chaos in a NaMLab Memristor

FRONTIERS IN NEUROSCIENCE

Ascoli, Alon; Demirkol, Ahmet S.; Tetzlaff, Ronald; Slesazeck, Stefan; Mikolajick, Thomas; Chua, Leon O.

被引用次数: 59

2021

RETRACTION: Φ memristor: Real memristor found (Retraction of Vol 125, art no 054504, 2019)

JOURNAL OF APPLIED PHYSICS

Wang, Frank Z.; Li, Ling; Shi, Luping; Wu, Huaqiang; Chua, Leon O.

被引用次数: 1

2021

Observation of stochastic resonance for weak periodic magnetic field signal using a chaotic system

COMMUNICATIONS IN NONLINEAR SCIENCE AND NUMERICAL SIMULATION

Silva, Iacyle G.; Korneta, Wojciech; Stavrinos, Stavros G.; Picos, Rodrigo; Chua, Leon O.

被引用次数: 16

2021

How to Build a Memristive Integrate-and-Fire Model for Spiking Neuronal Signal Generation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kang, Sung Mo; Choi, Donguk; Eshraghian, Jason K.; Zhou, Peng; Kim, Jieun; Kong, Bai-Sun; Zhu, Xiaojian; Demirkol, Ahmet Samil; Ascoli, Alon; Tetzlaff, Ronald; Lu, Wei D.; Chua, Leon O.

被引用次数: 27

2021

Neuromorphic Dynamics of Chua Corsage Memristor

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Jin, Peipei; Wang, Guangyi; Liang, Yan; Lu, Herbert Ho-Ching; Chua, Leon O.

被引用次数: 47

2021

Microtubules as Sub-Cellular Memristors

SCIENTIFIC REPORTS

Tuszynski, Jack A.; Friesen, Douglas; Freedman, Holly; Sbitnev, Valery I.; Kim, Hyongsuk; Santelices, Iara; Kalra, Aarat P.; Patel, Sahil D.; Shankar, Karthik; Chua, Leon O.

被引用次数: 36

2020

Experimental evaluation of the dynamic route map in the reset transition of memristive ReRAMs

CHAOS SOLITONS & FRACTALS

Maldonado, David; Gonzalez, Mireia B.; Campabadal, Francesca; Jimenez-Molinos, Francisco; Al Chawa, M. Moner; Stavrinos, Stavros G.; Roldan, Juan B.; Tetzlaff, Ronald; Picos, Rodrigo; Chua, Leon O.

被引用次数: 19

2020

Theoretical Foundations of Memristor Cellular Nonlinear Networks: A DRM2-Based Method to Design Memcomputers With Dynamic Memristors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ascoli, Alon; Tetzlaff, Ronald; Kang, Sung-Mo; Chua, Leon O.

被引用次数: 46

2020

Theoretical Foundations of Memristor Cellular Nonlinear Networks: Stability Analysis With Dynamic Memristors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ascoli, Alon; Messaris, Ioannis; Tetzlaff, Ronald; Chua, Leon O.

被引用次数: 41

2020

Theoretical Foundations of Memristor Cellular Nonlinear Networks: Memcomputing With Bistable-Like Memristors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Tetzlaff, Ronald; Ascoli, Alon; Messaris, Ioannis; Chua, Leon O.

被引用次数: 52

2020

RETRACTED: Φ memristor: Real memristor found (Retracted article. See vol. 129, 2021)

JOURNAL OF APPLIED PHYSICS

Wang, Frank Z.; Li, Ling; Shi, Luping; Wu, Huaqiang; Chua, Leon O.

被引用次数: 28

2019

Reply to comment on 'If it's pinched it's a memristor'

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Chua, Leon O.

被引用次数: 2

2019

Building cellular neural network templates with a hardware friendly learning algorithm

NEUROCOMPUTING

Adhikari, Shyam Prasad; Kim, Hyongsuk; Yang, Changju; Chua, Leon O.

被引用次数: 19

2018

Taming Spatiotemporal Chaos in Forced Memristive Arrays

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Buscarino, Arturo; Corradino, Claudia; Fortuna, Luigi; Chua, Leon O.

被引用次数: 8

2018

Scenario of the Birth of Hidden Attractors in the Chua Circuit

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Stankevich, Nataliya V.; Kuznetsov, Nikolay V.; Leonov, Gennady A.; Chua, Leon O.

被引用次数: 67

2017

A New Circuit for Emulating Memristors Using Inductive Coupling

IEEE ACCESS

Yu, Dongsheng; Zheng, Ciyang; Lu, Herbert Ho-Ching; Fernando, Tyrone; Chua, Leon O.

被引用次数: 39

2017

On Turing Instability in Nonhomogeneous Reaction-Diffusion CNN's

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Goras, Liviu; Ungureanu, Paul; Chua, Leon O.

被引用次数: 10	2017
Fractional memristor	
APPLIED PHYSICS LETTERS	
Wang, Frank Z.; Shi, Luping; Wu, Huaqiang; Helian, Na; Chua, Leon O.	
被引用次数: 25	2017
A Circuit-Based Neural Network with Hybrid Learning of Backpropagation and Random Weight Change Algorithms	
SENSORS	
Yang, Changju; Kim, Hyongsuk; Adhikari, Shyam Prasad; Chua, Leon O.	
被引用次数: 14	2017
Turing Patterns in Memristive Cellular Nonlinear Networks	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS	
Buscarino, Arturo; Corradino, Claudia; Fortuna, Luigi; Frasca, Mattia; Chua, Leon O.	
被引用次数: 29	2016
History Erase Effect in a Non-Volatile Memristor	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS	
Ascoli, Alon; Tetzlaff, Ronald; Chua, Leon O.; Strachan, John Paul; Williams, Richard Stanley	
被引用次数: 59	2016
The First Ever Real Bistable Memristors-Part I: Theoretical Insights on Local Fading Memory	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS	
Ascoli, Alon; Tetzlaff, Ronald; Chua, Leon O.	
被引用次数: 27	2016
The First Ever Real Bistable Memristors-Part II: Design and Analysis of a Local Fading Memory System	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS	
Ascoli, Alon; Tetzlaff, Ronald; Chua, Leon O.	
被引用次数: 28	2016
Parasitic Effects on Memristor Dynamics	
INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS	
Itoh, Makoto; Chua, Leon O.	
被引用次数: 20	2016
Hidden Bifurcations in the Multispiral Chua Attractor	
INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS	
Menacer, Tidjani; Lozi, Rene; Chua, Leon O.	
被引用次数: 25	2016
Generalized reconfigurable memristive dynamical system (MDS) for neuromorphic applications	

A Circuit-Based Learning Architecture for Multilayer Neural Networks With Memristor Bridge Synapses

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Adhikari, Shyam Prasad; Kim, Hyongsuk; Budhathoki, Ram Kaji; Yang, Changju; Chua, Leon O.

被引用次数: 141

2015

A memristor emulator as a replacement of a real memristor

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Yang, Changju; Choi, Hyuncheol; Park, Sedong; Sah, Maheshwar Pd; Kim, Hyongsuk; Chua, Leon O.

被引用次数: 47

2015

Associative Learning with Temporal Contiguity in a Memristive Circuit for Large-Scale Neuromorphic Networks

ADVANCED ELECTRONIC MATERIALS

Li, Yi; Xu, Lei; Zhong, Ying-Peng; Zhou, Ya-Xiong; Zhong, Shu-Jing; Hu, Yang-Zhi; Chua, Leon O.; Miao, Xiang-Shui

被引用次数: 63

2015

What are Memristor, Memcapacitor, and Meminductor?

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Yin, ZhenYu; Tian, Heng; Chen, GuanHua; Chua, Leon O.

被引用次数: 73

2015

Dynamic Behavior of Coupled Memristor Circuits

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Yu, Dongsheng; Lu, Herbert Ho-Ching; Liang, Yan; Fernando, Tyrone; Chua, Leon O.

被引用次数: 43

2015

A Theoretical Approach to Memristor Devices

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Corinto, Fernando; Civalieri, Pier Paolo; Chua, Leon O.

被引用次数: 65

2015

Cellular Memristive Dynamical Systems (CMDs)

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Bavandpour, Mohammad; Soleimani, Hamid; Bagheri-Shouraki, Saeed; Ahmadi, Arash; Abbott, Derek; Chua, Leon O.

被引用次数: 12

2014

Dynamics of Memristor Circuits

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Itoh, Makoto; Chua, Leon O.

被引用次数: 36

2014

Gas Discharge Lamps Are Volatile Memristors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Lin, Deyan; Hui, S. Y. Ron; Chua, Leon O.

被引用次数: 27

2014

Brains Are Made of Memristors

IEEE CIRCUITS AND SYSTEMS MAGAZINE

Sah, Maheshwar Pd; Kim, Hyongsuk; Chua, Leon O.

被引用次数: 137

2014

Powerful, Multifunctional Torsional Micromuscles Activated by Phase Transition

ADVANCED MATERIALS

Liu, Kai; Cheng, Chun; Suh, Joonki; Tang-Kong, Robert; Fu, Deyi; Lee, Sangwook; Zhou, Jian; Chua, Leon O.; Wu, Junqiao

被引用次数: 91

2014

A Universal Mutator for Transformations Among Memristor, Memcapacitor, and Meminductor

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Yu, Dongsheng; Liang, Yan; Lu, Herbert H. C.; Chua, Leon O.

被引用次数: 83

2014

Three Fingerprints of Memristor

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Adhikari, Shyam Prasad; Sah, Maheshwar Pd; Kim, Hyongsuk; Chua, Leon O.

被引用次数: 515

2013

The First Radios Were Made Using Memristors!

IEEE CIRCUITS AND SYSTEMS MAGAZINE

Gandhi, Gaurav; Aggarwal, Varun; Chua, Leon O.

被引用次数: 18

2013

Adaptive Neuromorphic Architecture (ANA)

NEURAL NETWORKS

Wang, Frank Zhigang; Chua, Leon O.; Yang, Xiao; Na Helian; Tetzlaff, Ronald; Schmidt, Torsten; Li, Caroline; Garcia Carrasco, Jose Manuel; Chen, Wanlong; Chu, Dominique

被引用次数: 42

2013

DUALITY OF MEMRISTOR CIRCUITS

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Itoh, Makoto; Chua, Leon O.

被引用次数: 42

2013

CANARDS FROM CHUA'S CIRCUIT

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Ginoux, Jean-Marc; Llibre, Jaume; Chua, Leon O.

被引用次数: 35

2013

Memristor Bridge Synapse-Based Neural Network and Its Learning

IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS

Adhikari, Shyam Prasad; Yang, Changju; Kim, Hyongsuk; Chua, Leon O.

被引用次数: 318

2012

Memristance drift avoidance with charge bouncing for memristor-based nonvolatile memories

JOURNAL OF THE KOREAN PHYSICAL SOCIETY

Adhikari, Shyam Prasad; Kim, Hyongsuk; Kong, Bai-Sun; Chua, Leon O.

被引用次数: 3

2012

Memristor Bridge Synapses

PROCEEDINGS OF THE IEEE

Kim, Hyongsuk; Sah, Maheshwar Pd.; Yang, Changju; Roska, Tamas; Chua, Leon O.

被引用次数: 231

2012

PHENOMENOLOGY OF RETAINED REFRACTORINESS: ON SEMI-MEMRISTIVE DISCRETE MEDIA

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Adamatzky, Andrew; Chua, Leon O.

被引用次数: 16

2012

Neural Synaptic Weighting With a Pulse-Based Memristor Circuit

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kim, Hyongsuk; Sah, Maheshwar Pd.; Yang, Changju; Roska, Tamas; Chua, Leon O.

被引用次数: 331

2012

Memristor Emulator for Memristor Circuit Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kim, Hyongsuk; Sah, Maheshwar Pd.; Yang, Changju; Cho, Seongik; Chua, Leon O.

被引用次数: 374

2012

MEMRISTOR HAMILTONIAN CIRCUITS

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Itoh, Makoto; Chua, Leon O.

被引用次数: 37

2011

TOPOLOGICAL ANALYSIS OF CHAOTIC SOLUTION OF A THREE-ELEMENT MEMRISTIVE CIRCUIT

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Ginoux, Jean-Marc; Letellier, Christophe; Chua, Leon O.

被引用次数: 11

2010

THE WOLFRAM MACHINE

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Fortuna, Luigi; Frasca, Mattia; Fiore, Angelo Sarra; Chua, Leon O.

被引用次数: 3

2010

AUTOASSOCIATIVE MEMORY CELLULAR NEURAL NETWORKS

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Itoh, Makoto; Chua, Leon O.

被引用次数: 8

2010

A NONLINEAR DYNAMICS PERSPECTIVE OF WOLFRAM'S NEW KIND OF SCIENCE. PART XIV: MORE BERNOULLI $\sigma\tau$ -SHIFT RULES

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Chua, Leon O.; Pazienza, Giovanni E.

被引用次数: 7

2010

A NONLINEAR DYNAMICS PERSPECTIVE OF WOLFRAM'S NEW KIND OF SCIENCE. PART XIII: BERNOULLI $\sigma\tau$ -SHIFT RULES

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Chua, Leon O.; Pazienza, Giovanni E.

被引用次数: 13

2010

SIMPLEST CHAOTIC CIRCUIT

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Muthuswamy, Bharathwaj; Chua, Leon O.

被引用次数: 420

2010

Circuit Elements With Memory: Memristors, Memcapacitors, and Meminductors

PROCEEDINGS OF THE IEEE

Di Ventra, Massimiliano; Pershin, Yuriy V.; Chua, Leon O.

被引用次数: 905

2009

DIFFERENCE EQUATIONS FOR CELLULAR AUTOMATA

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Itoh, Makoto; Chua, Leon O.

被引用次数: 4

2009

A NONLINEAR DYNAMICS PERSPECTIVE OF WOLFRAM'S NEW KIND OF SCIENCE. PART X: PERIOD-1 RULES

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Chua, Leon O.; Pazienza, Giovanni E.; Shin, Jinwook

被引用次数: 8

2009

A NONLINEAR DYNAMICS PERSPECTIVE OF WOLFRAM'S NEW KIND OF SCIENCE. PART XI: PERIOD-2 RULES

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Chua, Leon O.; Pazienza, Giovanni E.; Shin, Jinwook

被引用次数: 12

2009

A NONLINEAR DYNAMICS PERSPECTIVE OF WOLFRAM'S NEW KIND OF SCIENCE. PART XII: PERIOD-3, PERIOD-6, AND PERMUTIVE RULES

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Chua, Leon O.; Pazienza, Giovanni E.

被引用次数: 8

2009

MEMRISTOR CELLULAR AUTOMATA AND MEMRISTOR DISCRETE-TIME CELLULAR NEURAL NETWORKS

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Itoh, Makoto; Chua, Leon O.

被引用次数: 269

2009

A NONLINEAR DYNAMICS PERSPECTIVE OF WOLFRAM'S NEW KIND OF SCIENCE. PART IX: QUASI-ERGODICITY

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Chua, Leon O.; Pazienza, Giovanni Egidio; Orzo, Laszlo; Sbitnev, Valery I.; Shin, Jinwook

被引用次数: 18

2008

The four-element Chua's circuit

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Barboza, Ruy; Chua, Leon O.

被引用次数: 54

2008

SLOW INVARIANT MANIFOLDS AS CURVATURE OF THE FLOW OF DYNAMICAL SYSTEMS

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Ginoux, Jean-Marc; Rossetto, Bruno; Chua, Leon O.

被引用次数: 25

2008

Awakening dynamics via passive coupling and synchronization mechanism in oscillatory cellular neural/nonlinear networks

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Szatmari, Istvan; Chua, Leon O.

被引用次数: 14

2008

ISLES OF EDEN AND THE ZUK THEOREM IN $\mathbb{R}^d$

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Garay, Barnabas M.; Chua, Leon O.

被引用次数: 4

2008

APPLICATIONS OF LOCAL ACTIVITY THEORY OF CNN TO CONTROLLED COUPLED OREGONATOR SYSTEMS

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Min, Lequan; Meng, Yan; Chua, Leon O.

被引用次数: 1

2008

MEMRISTOR OSCILLATORS

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Itoh, Makoto; Chua, Leon O.

被引用次数: 859

2008

IMITATION OF VISUAL ILLUSIONS VIA OPENCV AND CNN

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Itoh, Makoto; Chua, Leon O.

被引用次数: 5

2008

A nonlinear dynamics perspective of Wolfram's new kind of science. part VII: Isles of Eden

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Chua, Leon O.; Guan, Junbiao; Sbitnev, Valery I.; Shin, Jinwook

被引用次数: 43

2007

A nonlinear dynamics perspective of Wolfram's new kind of science. Part VIII: More Isles of Eden

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Chua, Leon O.; Karacs, Chua Kristof; Sbitnev, Valery I.; Guan, Junbiao; Shin, Jinwook

被引用次数: 23

2007

How to represent continuous piecewise linear functions in closed form

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Lum, Robert; Chua, Leon O.

被引用次数: 3

2006

Nonlinear pattern classification associated with cellular neural networks-based dynamic programming

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Kim, HS; Oh, TW; Lim, MS; Chua, LO

被引用次数: 0

2005

Local activity is the origin of complexity

INTERNATIONAL JOURNAL OF BIFURCATION AND CHAOS

Chua, LO

被引用次数: 217

2005

High-performance Viterbi decoder with circularly connected 2-D CNN unilateral cell array

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

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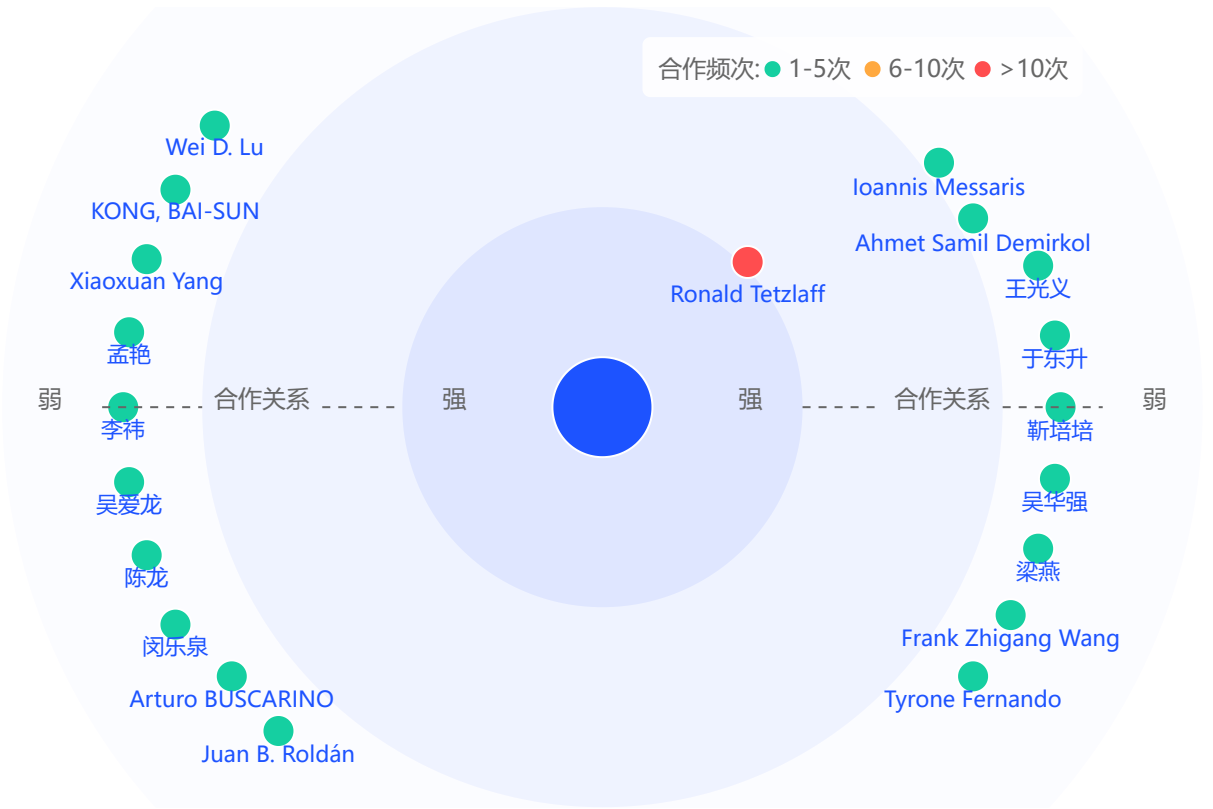
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